

Name: Savitha R

Reg.no: 192421368

C03-EXP5. Cross-Browser Compatible Interactive Webpage

Develop a webpage that uses feature detection and fallback techniques for unsupported browser features. Check support for localStorage, querySelector(), and suitable CSS features using @supports. Demonstrate how noncompliant or partially compliant browsers can be accommodated.

PROGRAM:

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
  <title>Cross-Browser Compatible Webpage</title>
```

```
<style>
```

```
  * {  
    box-sizing: border-box;  
  }
```

```
  body {  
    margin: 0;  
    font-family: Arial, sans-serif;  
    background: #ecfeff;  
    color: #164e63;  
  }
```

```
  header {  
    background: #155e75;
```

```
    color: #ffffff;
    padding: 35px 20px;
    text-align: center;
}
```

```
header h1 {
    margin: 0;
    font-size: 32px;
}
```

```
header p {
    color: #a5f3fc;
    margin-top: 10px;
}
```

```
.container {
    max-width: 950px;
    margin: 35px auto;
    padding: 20px;
}
```

```
.intro {
    background: #ffffff;
    padding: 25px;
    border-radius: 15px;
    border-left: 6px solid #0891b2;
    box-shadow: 0 5px 18px rgba(8, 145, 178, 0.15);
    margin-bottom: 25px;
}
```

```
.intro h2 {
    color: #0e7490;
```

```
    margin-top: 0;
}
```

```
.feature-container {
    display: flex;
    flex-wrap: wrap;
    gap: 20px;
}
```

```
@supports (display: grid) {
    .feature-container {
        display: grid;
        grid-template-columns: repeat(3, 1fr);
    }
}
```

```
.feature-card {
    flex: 1 1 250px;
    background: #ffffff;
    padding: 25px;
    border-radius: 15px;
    box-shadow: 0 5px 18px rgba(8, 145, 178, 0.12);
    border-top: 5px solid #f97316;
}
```

```
.feature-card h3 {
    color: #c2410c;
    margin-top: 0;
}
```

```
.status {
    padding: 12px;
```

```
border-radius: 8px;  
font-weight: bold;  
margin-top: 15px;  
}
```

```
.supported {  
  background: #dcfce7;  
  color: #166534;  
}
```

```
.unsupported {  
  background: #fee2e2;  
  color: #991b1b;  
}
```

```
.fallback {  
  background: #fef3c7;  
  color: #92400e;  
}
```

```
.storage-box {  
  margin-top: 25px;  
  background: #ffffff;  
  padding: 25px;  
  border-radius: 15px;  
  box-shadow: 0 5px 18px rgba(8, 145, 178, 0.12);s  
}
```

```
.storage-box h2 {  
  color: #0e7490;  
}
```

```
input {  
  width: 100%;  
  padding: 12px;  
  border: 2px solid #a5f3fc;  
  border-radius: 8px;  
  font-size: 15px;  
  margin: 10px 0;  
  outline: none;  
}
```

```
input:focus {  
  border-color: #0891b2;  
}
```

```
button {  
  padding: 12px 20px;  
  border: none;  
  border-radius: 8px;  
  background: #ea580c;  
  color: white;  
  font-size: 15px;  
  cursor: pointer;  
  margin: 5px;  
}
```

```
button:hover {  
  background: #c2410c;  
}
```

```
#message {  
  margin-top: 15px;  
  padding: 15px;
```

```
background: #cffee;
color: #155e75;
border-radius: 8px;
display: none;
}
```

```
.legacy {
  margin-top: 25px;
  padding: 20px;
  background: #fef2f2;
  border-radius: 12px;
  border-left: 5px solid #dc2626;
  color: #7f1d1d;
}
```

```
footer {
  margin-top: 30px;
  padding: 25px;
  text-align: center;
  background: #083344;
  color: #a5f3fc;
}
```

```
@media (max-width: 700px) {
  .feature-container {
    grid-template-columns: 1fr;
  }
}
```

```
header h1 {
  font-size: 25px;
}
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<header>
```

```
<h1>Cross-Browser Compatibility</h1>
```

```
<p>Feature Detection and Fallback Demonstration</p>
```

```
</header>
```

```
<div class="container">
```

```
<div class="intro">
```

```
<h2>Browser Compatibility Dashboard</h2>
```

```
<p>
```

```
  This webpage checks browser features before using them  
  and provides fallback methods when required features  
  are unavailable.
```

```
</p>
```

```
</div>
```

```
<div class="feature-container">
```

```
<div class="feature-card">
```

```
<h3>localStorage</h3>
```

```
<p>Checks whether browser storage is available.</p>
```

```
<div id="storageStatus" class="status">
```

```
  Checking...
```

```
</div>
```

```
</div>
```

```
<div class="feature-card">
```

<h3>querySelector()</h3>

<p>Checks support for modern DOM selection.</p>

<div id="queryStatus" class="status">

Checking...

</div>

</div>

<div class="feature-card">

<h3>CSS Grid</h3>

<p>Uses @supports to detect CSS Grid support.</p>

<div id="cssStatus" class="status">

Checking...

</div>

</div>

</div>

<div class="storage-box">

<h2>Browser Storage Test</h2>

<input

type="text"

id="userData"

placeholder="Enter a message to save"

>

<button id="saveButton">

Save Data

</button>

<button id="loadButton">

Load Data

</button>

<button id="clearButton">

Clear Data

</button>

<div id="message"></div>

</div>

<div class="legacy">

Fallback Support:

If localStorage is unavailable, the webpage uses temporary
in-memory storage. If querySelector() is unavailable,
getElementById() is used instead.

</div>

</div>

<footer>

Cross-Browser Compatible Interactive Webpage

</footer>

<script>

var querySupported =

typeof document.querySelector === "function";

var storageSupported = false;

try {

var testKey = "__browser_test__";

```
localStorage.setItem(testKey, "test");

localStorage.removeItem(testKey);

    storageSupported = true;
}
catch (error) {
    storageSupported = false;
}

var temporaryStorage = "";

function getElement(id) {
    if (querySupported) {
        return document.querySelector("#" + id);
    }

    return document.getElementById(id);
}

function setStatus(element, supported, normalText, fallbackText) {
    if (supported) {
        element.textContent = normalText;
        element.className = "status supported";
    }
    else {
        element.textContent = fallbackText;
        element.className = "status fallback";
    }
}
```

```
var storageStatus = getElement("storageStatus");
```

```
setStatus(  
    storageStatus,  
    storageSupported,  
    "Supported",  
    "Not Supported - Temporary Storage Used"  
);
```

```
var queryStatus = getElement("queryStatus");
```

```
setStatus(  
    queryStatus,  
    querySupported,  
    "Supported",  
    "Not Supported - getElementById() Used"  
);
```

```
var cssStatus = getElement("cssStatus");
```

```
if (  
    window.CSS &&  
    typeof CSS.supports === "function" &&  
    CSS.supports("display", "grid")  
) {  
    cssStatus.textContent = "Supported";  
    cssStatus.className = "status supported";  
}  
else {  
    cssStatus.textContent = "Not Supported - Flexbox Fallback Used";  
    cssStatus.className = "status fallback";  
}
```

```
var input = getElement("userData");
var message = getElement("message");
var saveButton = getElement("saveButton");
var loadButton = getElement("loadButton");
var clearButton = getElement("clearButton");

function showMessage(text) {
    message.textContent = text;
    message.style.display = "block";
}

function saveData() {
    if (input.value.trim() === "") {
        showMessage("Please enter some data.");
        return;
    }

    if (storageSupported) {
        localStorage.setItem("studentMessage", input.value);
        showMessage("Data saved using localStorage.");
    }
    else {
        temporaryStorage = input.value;
        showMessage("Data saved temporarily using fallback storage.");
    }
}

function loadData() {
    var savedData = "";

    if (storageSupported) {
```

```
    savedData =
        localStorage.getItem("studentMessage") || "";
    }
    else {
        savedData = temporaryStorage;
    }

    if (savedData !== "") {
        input.value = savedData;
        showMessage("Saved data loaded successfully.");
    }
    else {
        showMessage("No saved data found.");
    }
}

function clearData() {
    if (storageSupported) {
        localStorage.removeItem("studentMessage");
    }

    temporaryStorage = "";

    input.value = "";

    showMessage("Stored data has been cleared.");
}

if (typeof saveButton.addEventListener === "function") {

    saveButton.addEventListener("click", saveData);
```

```
loadButton.addEventListener("click", loadData);

clearButton.addEventListener("click", clearData);

}
else {

    saveButton.onclick = saveData;

    loadButton.onclick = loadData;

    clearButton.onclick = clearData;
}
</script>

</body>

</html>
```

Output:

Cross-Browser Compatibility

Feature Detection and Fallback Demonstration

Browser Compatibility Dashboard

This webpage checks browser features before using them and provides fallback methods when required features are unavailable.

localStorage

Checks whether browser storage is available.

Supported

querySelector()

Checks support for modern DOM selection.

Supported

CSS Grid

Uses @supports to detect CSS Grid support.

Supported

Browser Storage Test

Save Data

Load Data

Clear Data

Data saved using localStorage.

Fallback Support: If localStorage is unavailable, the webpage uses temporary in-memory storage. If querySelector() is unavailable, getElementById() is used instead.